

Embodiment Knowledge from Free Self-Play Video

LeRobot research — aloha sim (bimanual insertion) · VLA-JEPA recipe (arXiv 2602.10098) with an ACT student · June 10–12, 2026 · branch `proto/self-play-embodiment`

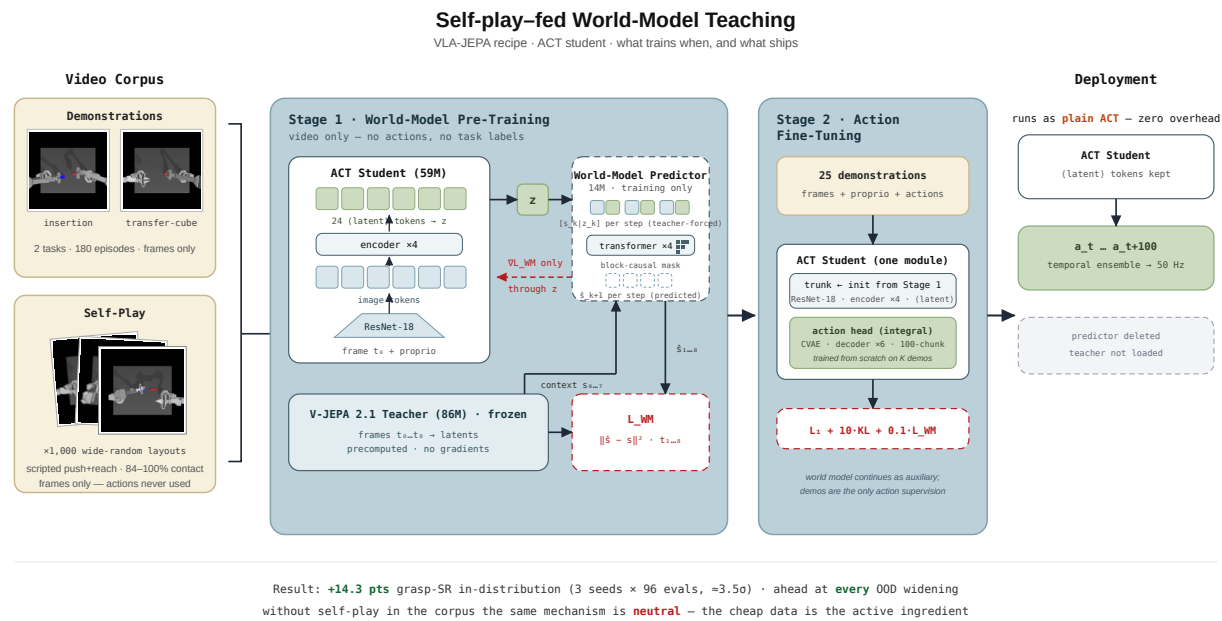
Question. Can a world-model module, trained on abundant *cheap* same-embodiment data (self-play), teach a small VLA so it needs fewer demos and generalizes better?

Headline (3 training seeds × 96 eval seeds). Feeding **1000 episodes of scripted self-play video** into the world-model corpus makes the K=25-demo policy **+14.3 pts better in-distribution (grasp rate 68.3% vs 54.0%, $\approx 3.5\sigma$, replicates per-seed)** and ahead at **every** out-of-distribution spawn widening (11/12 cells, sign-test $p \approx 0.003$). Demo-video-only stays neutral and self-play-only is $\approx$ *control* (50.3%†) — **the active ingredient is the union: target behavior + diversity**.

Terms & setup (read first)

term	meaning
task	AlohaInsertion (sim, bimanual): right arm grasps peg, left arm grasps socket, lift both, insert. Single top camera, 50 Hz.
K / demos	number of action-labeled demonstration episodes used in Stage 2 (e.g., K=25). Same fixed episodes for every arm.
Stage 1 / Stage 2	Stage 1 = world-model pre-training on VIDEO only (no actions). Stage 2 = action fine-tuning on the K demos (world model continues as a small auxiliary loss, then is deleted).
arms	control = no Stage 1, no WM (plain ACT from ImageNet init); WM(demos) = Stage-1 corpus is demo video only; WM(demos+SP) = + 1,000 self-play episodes; WM(SP-only) = self-play video only (zero demo video).
reward ladder	per episode, the max rung reached: 1 = both grippers contact their objects (dual grasp) · 2 = both lifted · 3 = peg touches socket · 4 = inserted (= true task success).
rung rates	We report plain-English rung rates: grasp (≥ 1), lift (≥ 2), align (≥ 3), insertion ($= 4$, the canonical success). Grasp rate is the headline because rung 4 is near-floor at this scale (0–20%) and statistically silent; grasp is the most populated rung AND the measured bottleneck (failures are grasp-execution failures). Full ladders are always reported. Calibration: on the canonical metric (SR=4), our K=25 control $\approx$ 8–21% — consistent with original ACT's $\approx$ 20% at 50 human demos.
shuffle-z gap	representation gate: how much world-model prediction degrades when each clip's latent tokens are swapped with another clip's. Large gap = the module carries clip-specific anticipation.
OOD f	evaluation with object spawn ranges widened $\times f$ beyond the demo distribution ($f=1$ in-distribution; $f=2$ doubles the range).
seeds × n	independent training seeds × evaluation episodes per cell (e.g., 3 seeds × 96 evals); adopted after measuring ± 14 -pt swings at $n=24$.

The validated recipe



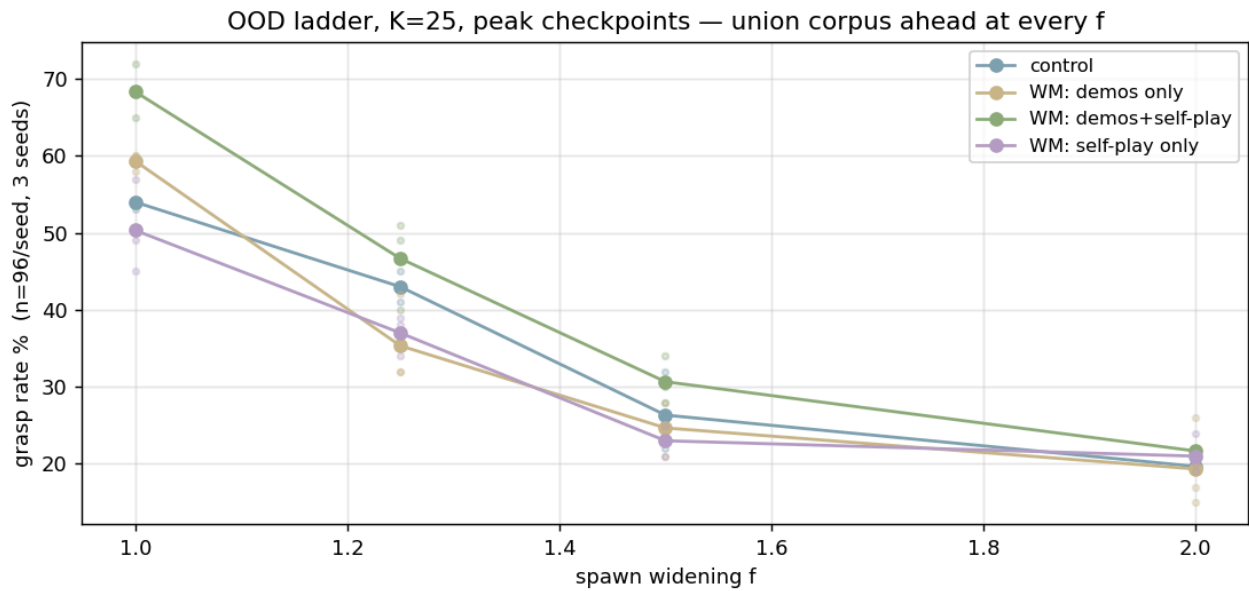
- **Self-play enters as video only** — the WM label is the future itself, so unscripted/imperfect behavior cannot corrupt training (actions never used).
- Faithful VLA-JEPA structure at 59M scale, verified against the paper + the upstream LeRobot port by an **executable self-test gate** (causality probes, leakage tests, mask-equality vs upstream, determinism) — training refuses to start if any check fails.

Result 1 — the module learns transferable anticipation (representation gate)

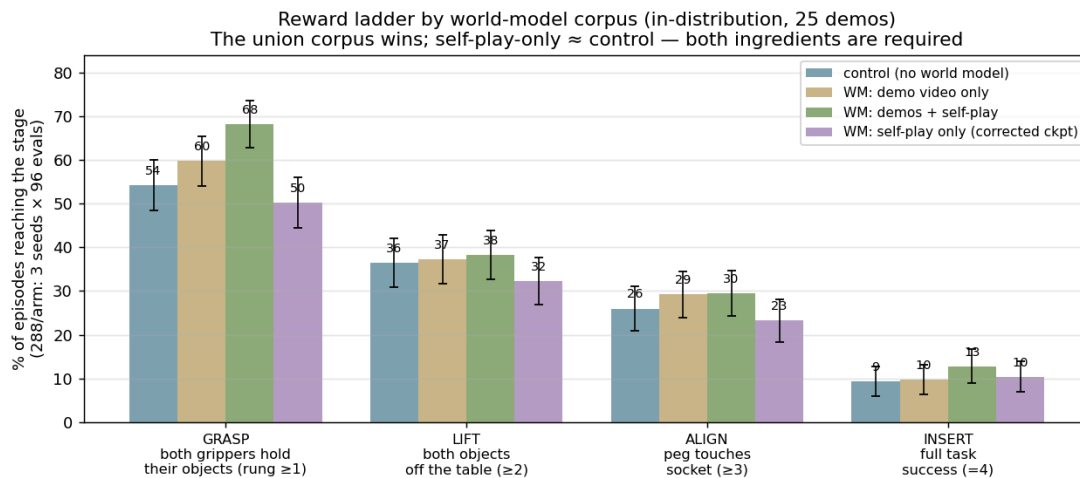
stage-1 corpus	wide-scene prediction (val/copy ↓)	wide-scene shuffle-z gap ↑
demos only	2.04 (worse than copying)	+1.3% (dead)
demos + self-play	0.574	+36.2%
self-play only	0.560	+34.1%

On **held-out wide layouts**, the demos-only module mispredicts and its latent-action channel carries nothing. With self-play in the corpus, anticipation transfers (3.5× better prediction; latent tokens carry scene-specific future info). In-distribution performance unchanged — the diversity costs nothing.

Result 2 — it converts to policy gains (3 seeds, n=96, peak protocol)



% episodes reaching grasp (rung ≥ 1), pooled	in-dist	1.25 $\times$ spawn	1.5 $\times$	2 $\times$
control (no WM)	54.0%	43.0%	26.3%	19.7%
WM, demos-only corpus	59.3%	35.3%	24.7%	19.3%
WM, demos+self-play	68.3%	46.7%	30.7%	21.7%
WM, self-play only†	50.3%	37.0%	23.0%	21.0%



- Honest framing: the win is a **higher level everywhere** + small absolute OOD edge — *not* a flatter relative decay (our pre-registered slope prediction failed).
- World model is **dropped at inference** — deployment cost identical to plain ACT.

Failure analysis — what still breaks (and what doesn't)

- **Reach generalizes:** grippers get within ~9 cm of displaced objects even at 2× widening (only ~1 cm worse than in-dist), failures included — measured across all 1,152 OOD episodes after first being spotted by eye in replay datasets.
- **Grasp/insert execution is the bottleneck**, in-distribution and OOD alike; secondary mode = late-episode instability past the demo-length tail. ⇒ next gains come from **grasp-diverse demos or grasp-capable play**, not more WM teaching.

What did NOT work (and saved us from scaling blind)

attempt	verdict
Bottleneck embodiment token on frozen encoder (Fork-2)	information-theoretically redundant for a competent policy
Fine-tuning the vision encoder with dynamics objectives (Fork-1/DynaMo)	injects body, evicts the object ; hurts policy at every demo count
WM teaching with task-redundant corpus (demos only)	neutral-to-harmful; dose-dependent interference at small scale
2B-scale VLA-JEPA reproduction	no-go: 8×A100 recipe; at our scale the in-dist effect (<1%) is below eval noise

Method practices the team can reuse

- **Paper-anchored code + executable self-test gate** — every claim in the paper maps to a runtime assertion; caught 3 silent deviations that had reversed an experimental conclusion.
- **Never compare arms at matched budget** — fixed steps can prove gains but not deficits; use per-arm stopping rules (val-plateau) and peak-over-checkpoints.
- **Closed-loop SR noise:** $n=24$ swings ± 14 pts across eval sets; $n \geq 96 \times \geq 3$ training seeds before believing any gap.
- **Visual failure review** (replayed eval episodes as GUI datasets) re-directed the OOD diagnosis from "reach lock-in" (wrong) to "grasp execution" (measured).

Recommended next steps

- **Option A — scale the validated recipe:** bigger student / real-ALOHA corpus arm / more & richer self-play. Evidence basis now exists.
- **Option B — attack the measured bottleneck:** grasp-diverse demonstrations or grasp-capable play (feeds both representation and action channels).
- **Artifacts:** self-play dataset: huggingface.co/datasets/thewisp/aloha_sim_selfplay_wide_1k (1,000 eps; card documents the 0–969 train / 970–999 val split) · code + full experimental record: github.com/TheWisp/lerobot · [proto/self-play-embodiment](#) · [research/self_play_embodiment/](#) (FINDINGS.md, selftest-gated scripts, per-seed results, figures).

† corrected row: the SP-only arm initially scored 47.7% under a checkpoint selected by a necessarily-different stopping gate; re-running from the later checkpoint (full protocol, 3 seeds) gives 50.3% — checkpoint-selection sensitivity is a measured ± 3 –8 pts. Its trunk also only ever practiced anticipation from episode starts (150-frame episodes vs the 128-frame window) — fixable with longer collection. · Generated 2026-06-12 · experiments: aloha sim AlohaInsertion-v0, ACT-style student (59M) + frozen V-JEPA 2.1 teacher · 3 training seeds × 96 paired eval seeds per OOD cell · single RTX 5090